Abstract Algebra - Stockholm University 2025

1. Groups

1.1. Conjugation, centralizer, center, normalizer

For $g \in G$, the conjugation map $c_g(x) = gxg^{-1}$ is an automorphism of G (an *inner* automorphism). For a subset $S \subseteq G$, the centralizer is

$$C_G(S) = \{\, g \in G : \ gs = sg \ \text{ for all } s \in S \,\}; \quad C_G(g) := C_G(\{g\}).$$

The center is

$$Z(G) = C_G(G) = \{g \in G : \ gx = xg \ \forall x \in G\}$$

(always a characteristic, hence normal, subgroup). For a subgroup $H \le G$, the normalizer is

$$N_G(H) = \{\, g \in G : \ gHg^{-1} = H \,\},$$

the largest subgroup of G in which H is normal (so $H \trianglelefteq N_G(H)$). One always has $C_G(H) \le N_G(H)$. The number of conjugates of H in G equals $[G : N_G(H)]$.

1.2. Normal subgroups and quotients

A subgroup $N \le G$ is normal (write $N \trianglelefteq G$) if $gNg^{-1} = N$ for all $g \in G$. Equivalently, every left coset equals the corresponding right coset: $gN = Ng$. If $N \trianglelefteq G$, the set of cosets $G/N = \{gN : g \in G\}$ is a group with $(gN)(hN) = (gh)N$.

1.3. Cosets and index

For $H \le G$ and $g \in G$, the left (resp. right) coset is $gH = \{gh : h \in H\}$ (resp. $Hg = \{hg : h \in H\}$). Left cosets are either disjoint or equal and they partition G. Each left coset gH has $|H|$ elements via the bijection $H \to gH$, $h \mapsto gh$. The *index* of H in G is the number of left cosets, denoted $[G\!:\!H]$.

1.4. Lagrange's Theorem

If G is finite and $H \le G$, then

$$|G| \ = \ [G\!:\!H] \cdot |H|.$$

Proof sketch. G is the disjoint union of $[G\!:\!H]$ left cosets, each of cardinality $|H|$.

Consequences

- $|H| \mid |G|$ for every $H \le G$.
- For $g \in G$, the order $|g|$ divides $|G|$; hence $g^{|G|} = e$.
- Index multiplicativity: for $K \le H \le G$ (with G finite), $[G\!:\!K] = [G\!:\!H]\,[H\!:\!K]$.
- If $[G\!:\!H] = 2$, then $H \trianglelefteq G$ (there are only two cosets, so gHg^{-1} must be H).

Groups of prime order are cyclic. If $|G| = p$ (prime) and $g \in G$, $g \neq e$, then $|\langle g\rangle| = |g| \mid p$ by Lagrange, so $|g| = p$ and $\langle g\rangle = G$. Hence G is cyclic and $G \cong \mathbb{Z}/p\mathbb{Z}$.

1.5. Isomorphism theorems

(1) **First isomorphism theorem.** For $\varphi : G \to H$ a homomorphism,

$$G/\ker\varphi \;\cong\; \operatorname{im}\varphi.$$

(2) **Second isomorphism theorem.** For $A \le G$ and $N \trianglelefteq G$,

$$A \cap N \trianglelefteq A, \qquad AN \le G, \qquad AN/N \;\cong\; A/(A \cap N).$$

(3) **Third isomorphism theorem.** If $K \le H \trianglelefteq G$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \;\cong\; G/H.$$

(4) **Correspondence (lattice) theorem.** For $N \trianglelefteq G$, the map

$$\{\,\text{subgroups } H \mid N \le H \le G\,\} \;\longleftrightarrow\; \{\,\text{subgroups of } G/N\,\}, \quad H \mapsto H/N$$

is a bijection preserving inclusion and index; it carries normal subgroups $H \trianglelefteq G$ with $N \le H$ to normal subgroups of G/N.

2. Group actions

2.1. Actions on cosets and by conjugation

- For $H \le G$, the action $G \curvearrowright G/H$ by left multiplication is transitive, with stabilizer of aH equal to aHa^{-1}.
- The action $G \curvearrowright G$ by conjugation, $g \cdot x = gxg^{-1}$, partitions G into conjugacy classes. The stabilizer of x is its centralizer $C_G(x)$, and $|\mathrm{Orb}(x)| = [G : C_G(x)]$.

2.2. Cayley's Theorem

Every group G is isomorphic to a subgroup H of the symmetric group $S_{|G|}$:

$$G \cong H \le S_{|G|}.$$

2.3. Orbit–stabilizer

For any action $G \curvearrowright X$ and any $x \in X$ there is a canonical bijection

$$G/G_x \xrightarrow{\sim} \mathrm{Orb}(x), \qquad gG_x \longmapsto g \cdot x.$$

In particular, if G is finite then

$$|\mathrm{Orb}(x)| = [G : G_x] \quad \text{and} \quad |G| = |\mathrm{Orb}(x)| \cdot |G_x|.$$

2.4. Class Equation

For a finite group G, let $Z(G)$ be the center of G. Then

$$|G| \;=\; |Z(G)| \;+\; \sum_{i=1}^{r} [G : C_G(g_i)],$$

where $g_1, \ldots, g_r$ are representatives of the conjugacy classes not contained in $Z(G)$.

2.5. Conjugacy in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type (i.e. the same cycle lengths with the same multiplicities).

$$\sigma, \tau \in S_n \quad \text{are conjugate} \quad \Longleftrightarrow \quad \text{cycle-type}(\sigma) = \text{cycle-type}(\tau).$$

Hence, the conjugacy classes of S_n correspond bijectively to partitions of n.

2.6. Automorphisms of cyclic groups

For $n \geq 1$, the automorphism group of the cyclic group $C_n \cong \mathbb{Z}/n\mathbb{Z}$ is

$$\mathrm{Aut}(C_n) \;\cong\; (\mathbb{Z}/n\mathbb{Z})^{\times},$$

hence $|\mathrm{Aut}(C_n)| = \varphi(n)$. Here $\varphi(n)$ is *Euler's totient function*:

$$\varphi(n) = \left|(\mathbb{Z}/n\mathbb{Z})^{\times}\right| = \#\{\, 1 \leq k \leq n : \gcd(k, n) = 1 \,\}.$$

Equivalently, if $n = \prod p_i^{e_i}$ then

$$\varphi(n) = n \prod_i \Big(1 - \frac{1}{p_i}\Big) = \prod_i \big(p_i^{e_i} - p_i^{e_i - 1}\big).$$

3. Sylows Theorem

4. Rings

4.1. Chinese Remainder Theorem

Let R be a commutative ring with 1 and let $A_1, \ldots, A_k \trianglelefteq R$ be ideals. If the A_i are pairwise comaximal ($A_i + A_j = R$ for $i \neq j$), then the natural map

$$\phi : R \longrightarrow \prod_{i=1}^{k} R/A_i, \qquad r \mapsto (r + A_i)_i$$

is surjective with kernel $\bigcap_{i=1}^{k} A_i = \prod_{i=1}^{k} A_i$. Hence

$$R/\big(\textstyle\prod_{i=1}^{k} A_i\big) \;\cong\; \textstyle\prod_{i=1}^{k} R/A_i.$$

5. Integral -, Unique Factorization - & Principle Ideal Domains

5.1. Gauss's Lemma

Let R be a UFD with field of fractions F, and let $p(x) \in R[x]$. If p is reducible in $F[x]$, then p is reducible in $R[x]$. (Equivalently: if p is irreducible in $R[x]$, then it is irreducible in $F[x]$.)

5.2. Eisenstein's Criterion

Let R be an integral domain and $P \subset R$ a prime ideal. Consider

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in R[x].$$

If $a_{n-1}, \ldots, a_0 \in P$ and $a_0 \notin P^2$, then $f(x)$ is irreducible in $R[x]$.

Corollary ($\mathbb{Z}[x]$). Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$. If there exists a prime p such that $p \mid a_i$ for $i = 0, \ldots, n-1$ and $p^2 \nmid a_0$, then f is irreducible in $\mathbb{Z}[x]$ (hence in $\mathbb{Q}[x]$ by Gauss).

6. Field Extensions